Short Syllabus

BCSE418L Explainable Artificial Intelligence (2-0-0-2)

XAI - Fundamentals, Categorizations, Taxonomy, Machine Learning Interpretability - Causal Model Induction - Causality learning ; Interpretability - Logistic regression, Generalized Additive Models, Decision Tree - Neural network interpretation ; Attention Mechanisms - Modular Networks - Feature Identification - Visualization -Feature, Deep - Sensitivity analysis; models - AHE, PHE, BETA ; XAI Techniques - LIME, SHAP, DiCE, LRP; Metrics to evaluate XAI - Evaluating AI system ; Medical diagnosis - Sales predictions on the house sale.

Course code	Course Title		L	T	P	C
BCSE418L	**Explainable Artificial Intelligence**		2	0	0	2
Pre-requisite	**NIL**	**Syllabus version**				
		1.0				

Course Objectives

1. To familiarize concepts related to Explainable Artificial Intelligence (XAI) and interpretable methods, with emphasis on how to build a trustworthy AI system.
2. To understand the performance of a machine learning model and its ability to produce explainable and interpret able predictions.

Course Outcomes

At the end of this course, student will be able to:

1. Understand the methods and terminologies involved in Explainable AI,
2. Differentiate the methods used in XAI and apply suitable XAI Models or approaches for given application.
3. Design and develop XAI use cases for real time applications.
4. Design of test procedures to assess the efficiency of the developed model

Module:1	**Introduction to Explainable Artificial Intelligence**	**4 hours**
Fundamentals of XAI - Categorization of XAI - Taxonomy of XAI methods for Machine Learning - Machine Learning Interpretability - Causal Model Induction - Causality learning - XAI techniques and limitations.		
Module:2	**Interpretability**	**5 hours**
Difference between Interpretability and Explainability - Interpretability methods to explain Black-Box Model - Scope of Interpretability - Apply interpretability on Regression, Logistic regression, Generalized Additive Models, Decision Tree - Neural network interpretation - Evaluation of Interpretability		
Module:3	**Deep Explanation**	**4 hours**
Attention Mechanisms - Modular Networks - Feature Identification - Learn to Explain - Feature Visualization - Deep Visualization- gradcam and Activation maps - Sensitivity analysis -		
Module:4	**XAI Models**	**5 hours**
Ante-hoc Explainability (AHE) models - Post-hoc Explainability (PHE) models - Interactive Machine Learning (IML) - Black Box Explanation through Transparent Approximation (BETA) models - Hybrid Models.		
Module:5	**XAI Methods**	**5 hours**
XAI Techniques - Local Interpretable Model-Agnostic Explanations (LIME) - Understanding Mathematical representation of LIME - Shapley Additive exPlanations (SHAP) - Diverse Counterfactual Explanations (DiCE) - Layer wise Relevance Propagation (LRP).		
Module:6	**Trust and acceptance**	**3 hours**
Metrics to evaluate XAI, Trustworthy Explainability Acceptance, Power Quality Disturbance (PQD) classification, Methods for measuring human intelligence. Evaluating AI system.		
Module:7	**Building Trustworthy Model with Explainable AI**	**3 hours**
Medical diagnosis- Making AI Decisions Trustworthy for Physicians and Patients – Sales predictions on the house sale.		
Module:8	**Recent Trends**	**1 hours**
	Total Lecture hours:	**30 Hours**
Text Book(s)		

1.	Molnar, Christoph. "Interpretable machine learning. A Guide for Making Black Box Models Explainable", 2019. https://christophm.github.io/interpretable-ml-book/.
2	Explainable Artificial Intelligence: An Introduction to Interpretable Machine Learning, Uday Kamath: John Liu, Springer, ISBN 9783030833558
Reference Books	
1.	Tim Miller Explanation in Artificial Intelligence: Insight from Social Science, https://arxiv.org/abs/1706.07269
2.	A Guide for making black-box machine learning models https://christophm.github.io/interpretable-ml-book/
3	Explainable AI: A Review of Machine Learning Interpretability Methods https://www.mdpi.com/1099-4300/23/1/18
4	Lötsch, J.; Kringel, D.; Ultsch, A. Explainable Artificial Intelligence (XAI) in Biomedicine: Making AI Decisions Trustworthy for Physicians and Patients. BioMedInformatics 2022, 2, 1-17. https://doi.org/10.3390/biomedinformatics2010001
Mode of Evaluation: CAT / Written Assignment / Quiz / FAT	

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